

ONCOLOGY PRECISION MEDICINE PLATFORM

STAGE 04 — INTEGRATION SUBSYSTEM: FINALIZED DECISIONS

Project: Personalized Precision Medicine for Oncology Treatment Optimization

Target: Multimodal Integration of Stage 01 (ML), Stage 02 (DL), and Stage 03 (NLP)

Status: **APPROVED BY USER** | Protocol: **STRICT READ-ONLY UPSTREAM**

1. Executive Verdict & Plan Status

The Stage 04 multimodal integration plan has been formally reviewed and APPROVED.

Execution will construct an isolated, read-only integration layer inside STAGE_04_INTEGRATION/.

Components Approved:

- Architecture: APPROVED
- Folder Structure: APPROVED
- Adapters (S1/S2/S3): APPROVED
- Patient Alignment: APPROVED
- Multimodal Fusion: APPROVED
- Safety Engine: APPROVED
- Explainability API: APPROVED
- Quality Gate (25-pt): APPROVED
- Unit Test Suite (11): APPROVED
- Audit & Manifests: APPROVED

2. Finalized Core Decisions & Corrections

CORRECTION 1: Renormalized Fusion Formula (Available Modalities)

The multimodal fusion formula must explicitly sum and normalize strictly over available modalities M_{avail} :

$$S_{integrated} = \sum_{m \in M_{avail}} (w_m \times P_m(HIGH)) / \sum_{m \in M_{avail}} w_m$$

- If NLP is missing: $S_{integrated} = (w_{ML} \times P_{ML} + w_{DL} \times P_{DL}) / (w_{ML} + w_{DL})$
- Missing modalities are NEVER treated as zero risk; remaining active weights renormalize to 1.0.
- Integration status is explicitly labeled: FULL_MULTIMODAL, PARTIAL_MULTIMODAL, or SINGLE_MODALITY.

CORRECTION 2: Truthful Patient Alignment & Zero Fabrication

- Raw ID Reality: Stage 01/02 use TCGA IDs (TCGA-XX-XXXX), while Stage 03 uses synthetic IDs (SYNTH_PAT_XXXX).
- Exact String Overlap: S1 & S2 share 157 patients (24 in test split); S1/S2 and S3 share 0 raw string IDs.
- Non-Fabrication Rule: The alignment report must honestly state actual exact match overlap (0 triple raw overlap).
- Benchmarking Bridge: To validate the full 3-modality pipeline under audit, an explicit, read-only synthetic crosswalk table (synthetic_benchmark_crosswalk.json) links 24 test patients across matching cancer types.
- Both raw exact match statistics and benchmark cohort results are reported transparently.

MANDATORY SYNTHETIC DATA & REGULATORY DISCLAIMER

Synthetic research integration only. This system is not clinically validated and must not be interpreted as evidence of clinical efficacy, diagnostic performance, or treatment recommendation.

All models, data, and decision rules evaluated herein represent a research decision-support prototype developed on synthetic oncology data.

3. Sequential 24-Step Implementation Sequence

STEP 01: Discover upstream S1/S2/S3 artifacts	STEP 13: Execute 4-rule clinical safety engine
STEP 02: Validate Stage 01 ML artifacts & models	STEP 14: Calculate cross-stage model agreement taxonomy
STEP 03: Validate Stage 02 DL artifacts & checkpoints	STEP 15: Generate patient explainability decompositions
STEP 04: Validate Stage 03 NLP predictions & NER	STEP 16: Generate batch predictions (CSV & JSON)
STEP 05: Load and catalog all patient identifiers	STEP 17: Generate 10 publication-quality 300 DPI figures
STEP 06: Execute exact string & benchmark alignment	STEP 18: Generate immutable structured audit logs
STEP 07: Catalog and report unmatched records	STEP 19: Execute reproducibility audit & config hashing
STEP 08: Aggregate longitudinal notes & multi-image sets	STEP 20: Execute 11-module unit test suite
STEP 09: Normalize confidence & validate probability bounds	STEP 21: Execute automated 25-point quality gate
STEP 10: Validate standardized output schemas	STEP 22: Generate 10 markdown analytical reports
STEP 11: Execute dynamic multimodal fusion engine	STEP 23: Generate final handover master report
STEP 12: Calculate calibrated integrated risk scores	STEP 24: Print official completion terminal banner

4. Automated 25-Point Integration Quality Gate

Check 01: S1 Outputs Exist	Check 10: DL Probabilities Valid	Check 19: Patient Aggregation Valid
Check 02: S2 Outputs Exist	Check 11: NLP Probabilities Valid	Check 20: Predictions Generated
Check 03: S3 Outputs Exist	Check 12: Probability Sums ≈ 1.0	Check 21: Explanations Generated
Check 04: All Artifacts Load	Check 13: No NaN Integrated Scores	Check 22: Audit Log Generated
Check 05: Patient IDs Valid	Check 14: No Inf Integrated Scores	Check 23: Unit Tests Pass (11/11)
Check 06: Duplicate IDs Detected	Check 15: Fusion Weights Sum = 1.0	Check 24: Reproducibility Verified
Check 07: Patient Alignment Complete	Check 16: Missing Modalities Handled	Check 25: Master Report Generated
Check 08: Unmatched Records Logged	Check 17: Safety Rules Executed	
Check 09: ML Probabilities Valid	Check 18: Model Agreement Calculated	

5. Expected Final Deliverables & Handover

- Codebase: STAGE_04_INTEGRATION/ (config, adapters, alignment, normalization, fusion, safety, explainability, inference, audit, validation, reproducibility, tests, outputs, visualizations, reports).
- Master Runner: STAGE_04_INTEGRATION/run_integration_engineer.py
- Master Validation Gate: STAGE_04_INTEGRATION/validate_integration.py
- Public Inference API: STAGE_04_INTEGRATION/inference.py (analyze_patient, analyze_multimodal_record)
- Batch Predictions: integrated_patient_predictions.csv and integrated_patient_predictions.json
- Patient Alignment: patient_alignment.csv and patient_alignment_report.json
- 10 Comprehensive Markdown Reports in reports/ including integration_engineering_master_report.md
- 10 Publication-Quality 300 DPI Figures in visualizations/

ENGINEERING HANDOVER SPECIFICATION

Subsystem: STAGE 04 INTEGRATION | Role: Integration Engineer | Upstream Stages: 01 (ML), 02 (DL), 03 (NLP)

Next Handover Target: SYSTEM / APPLICATION / DEPLOYMENT ENGINEER